

M3331 Codec Control Board Quick Start Guide

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1 OVERVIEW

The NuMaker-M3334KI_V1.0 is the evaluation board for M3334KIGAE. This board integrates an NAU88C22 codec, a 3.5 mm audio jack, and a microphone, making it suitable for audio development and evaluation.

To accelerate user development with the NAU88C22, firmware is provided to enable the NuMaker-M3334KI_V1.0 to function as a USB control board. With this firmware, users can:

- Access and configure NAU88C22 registers via I2C.
- Perform audio playback and recording via I2S.

In combination with the NuVotonAudioGUI, users can quickly set up and operate the NAU88C22 on their PCs with a graphical interface.

This document provides:

- Instructions for flashing the firmware.
- Guidelines for connecting the NuMaker-M3334KI_V1.0 to a PC.
- A basic user guide for NuvotonAudioGUI.

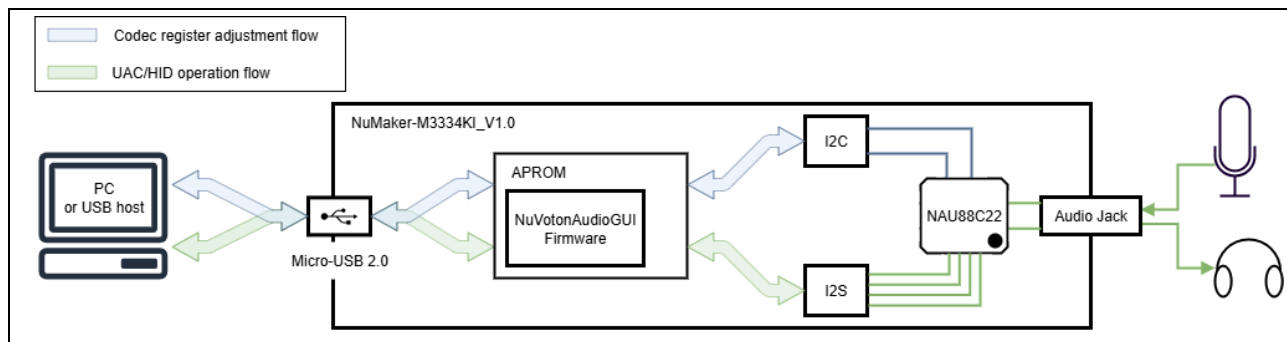
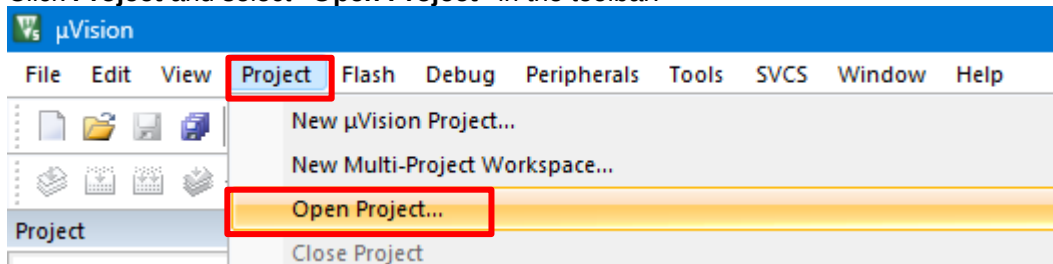


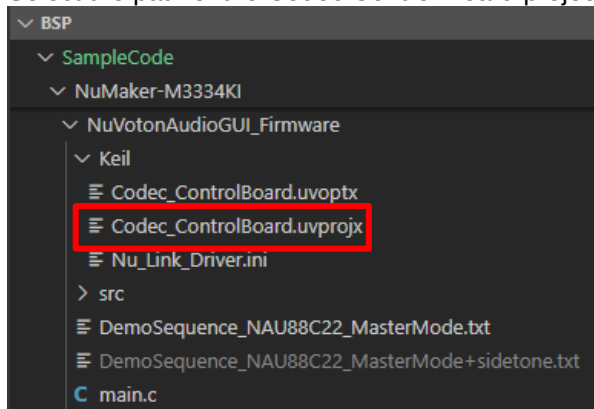
Figure 1-1 Block Diagram

2 GET STARTED WITH A KEIL PROJECT

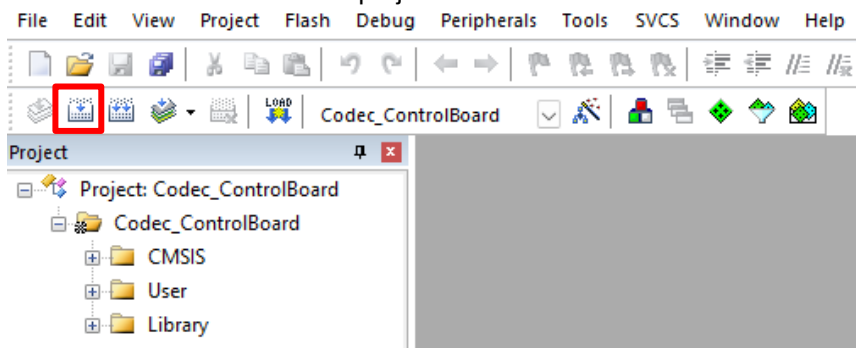
1. Click **Project** and select "Open Project" in the toolbar.



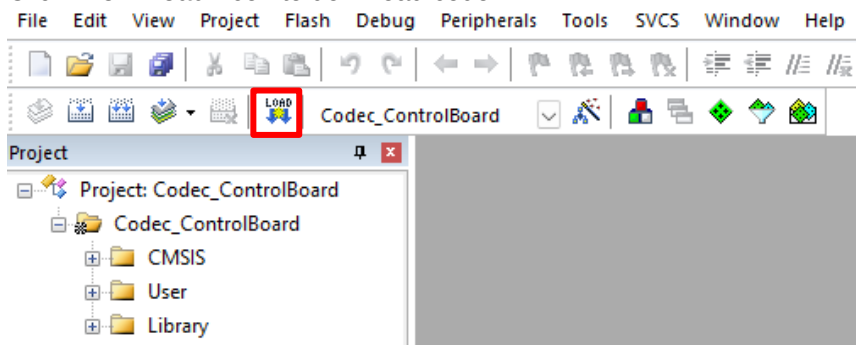
2. Select the path of the Codec Control Board project as below.



3. Click "Build" icon to build the project.



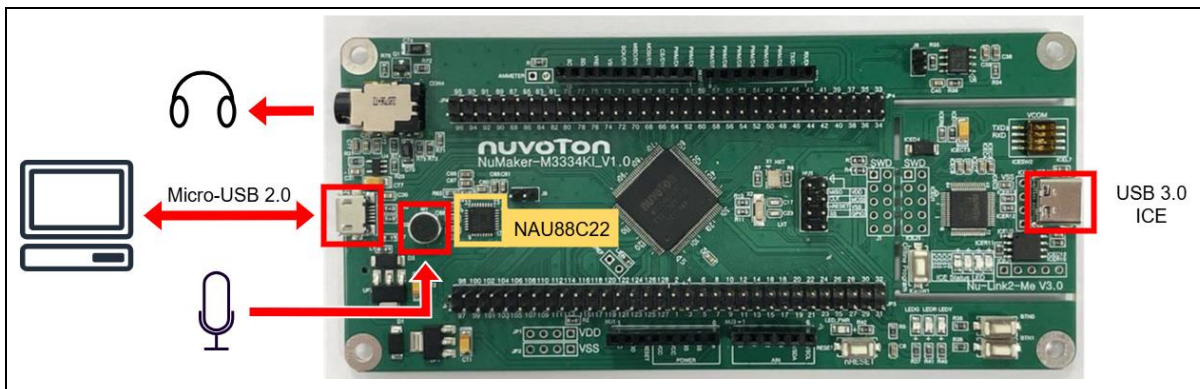
4. Click "Download" icon to download code.



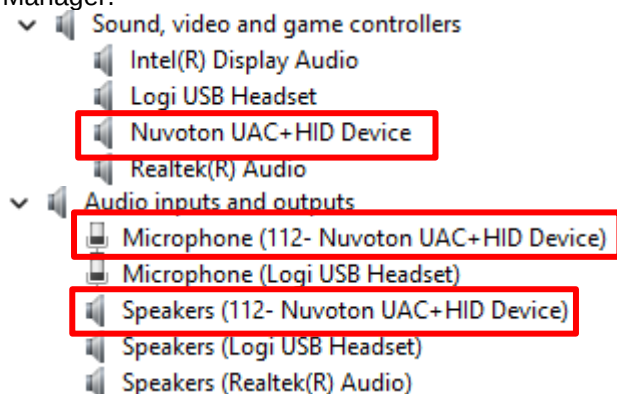
3 HARDWARE CHECK AND CONNECTION

Before using NuvotonAudioGUI, please confirm the hardware configured as follows before connecting to a Windows based PC.

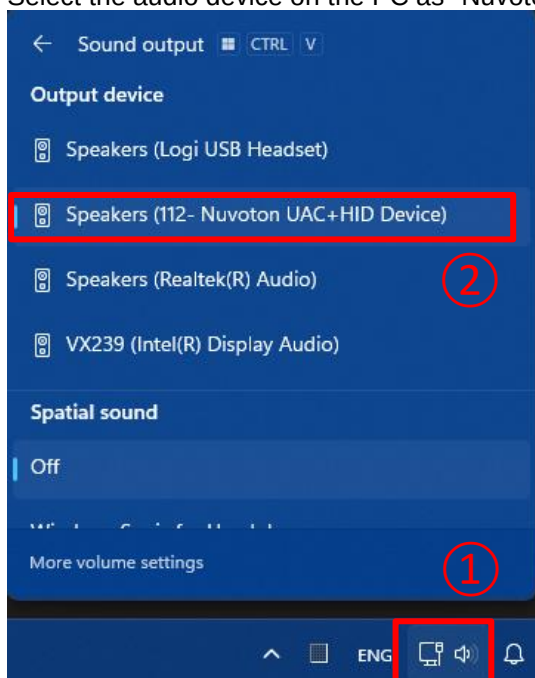
1. Plug in the Micro-USB 2.0 connector on NuMaker-M3334KI_V1.0.



2. After successfully recognizing the device, "Nuvoton UAC+HID Device" can be seen in Device Manager.



3. Select the audio device on the PC as "Nuvoton UAC+HID Device".

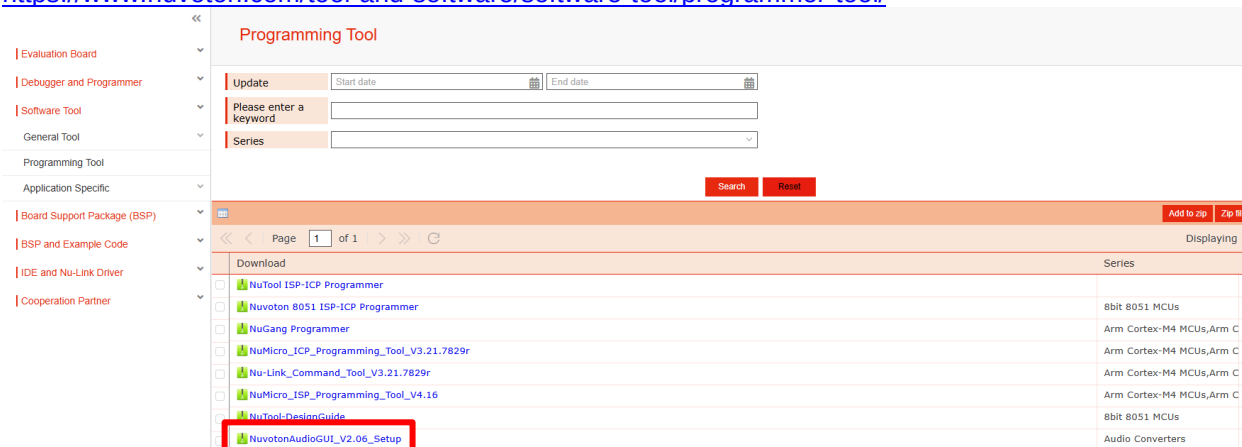


4 INSTALL GUI

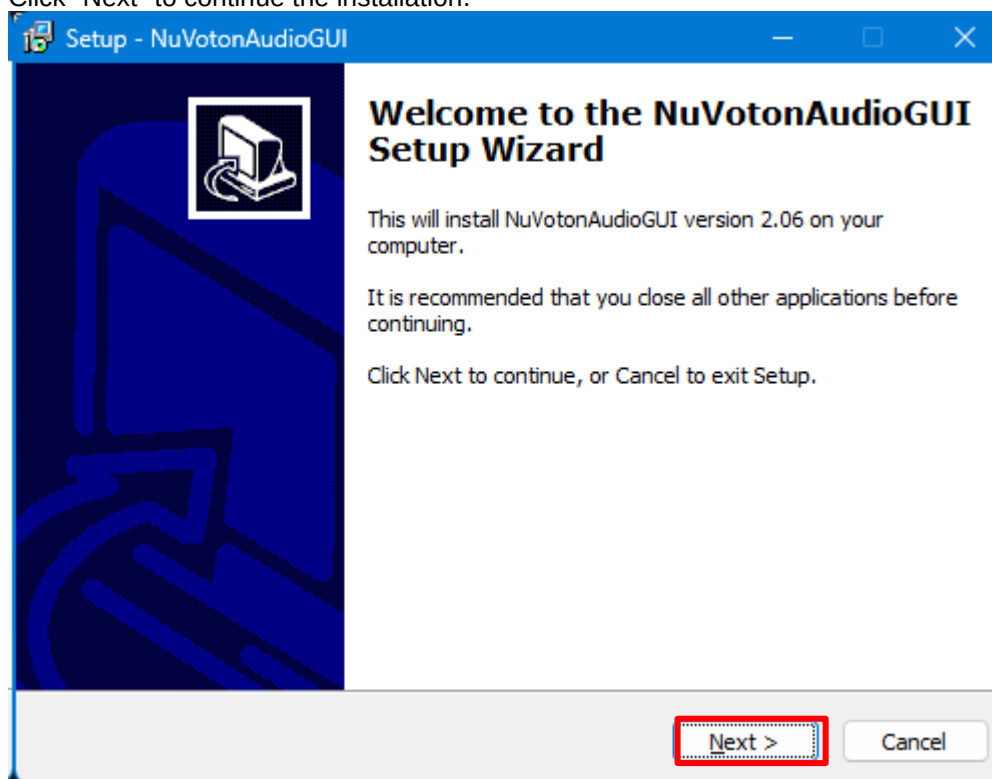
1. Visit Nuvoton Website.

Download NuVotonAudioGUI software.

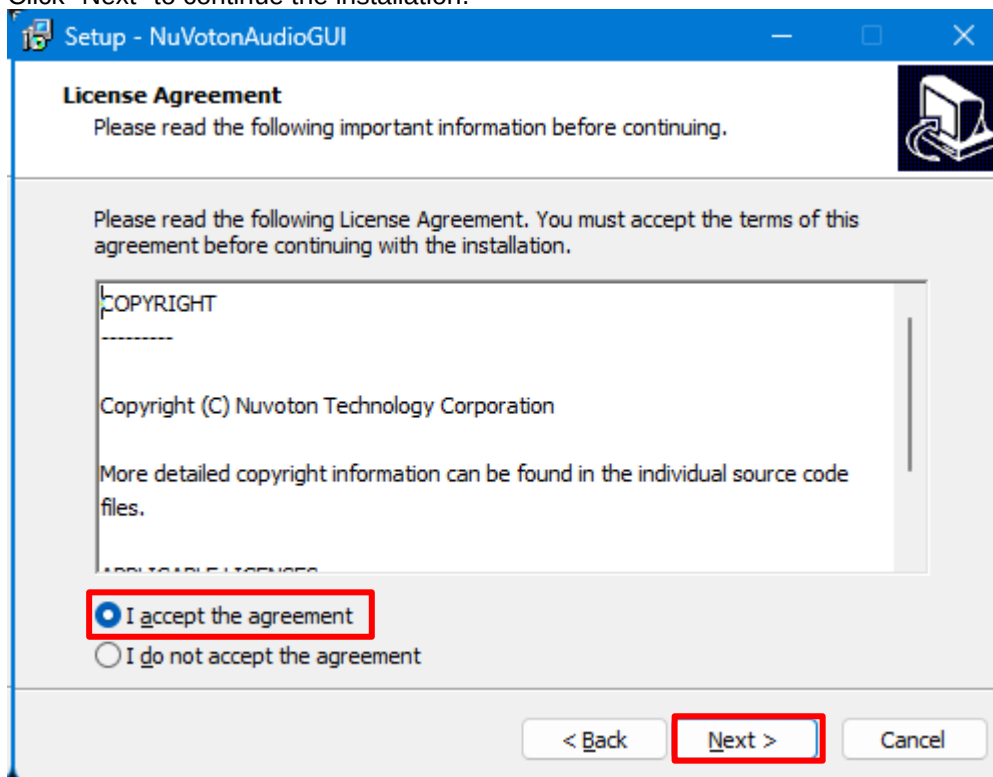
<https://www.nuvoton.com/tool-and-software/software-tool/programmer-tool/>



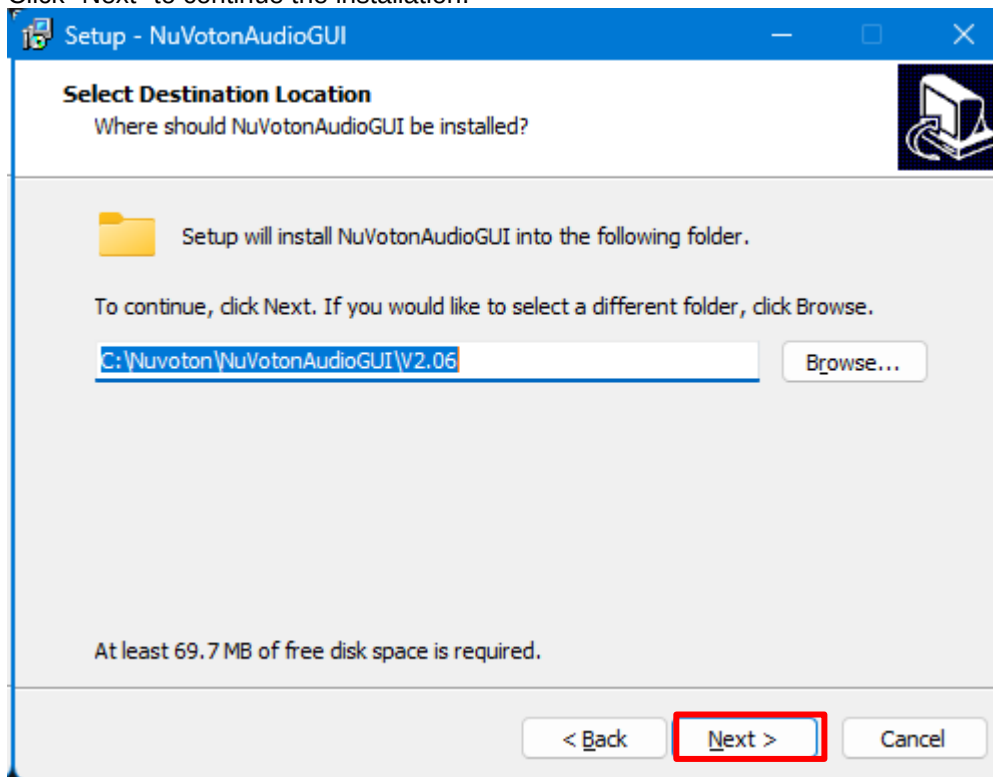
2. Run the NuVotonAudioGUI installer.
3. Click "Next" to continue the installation.



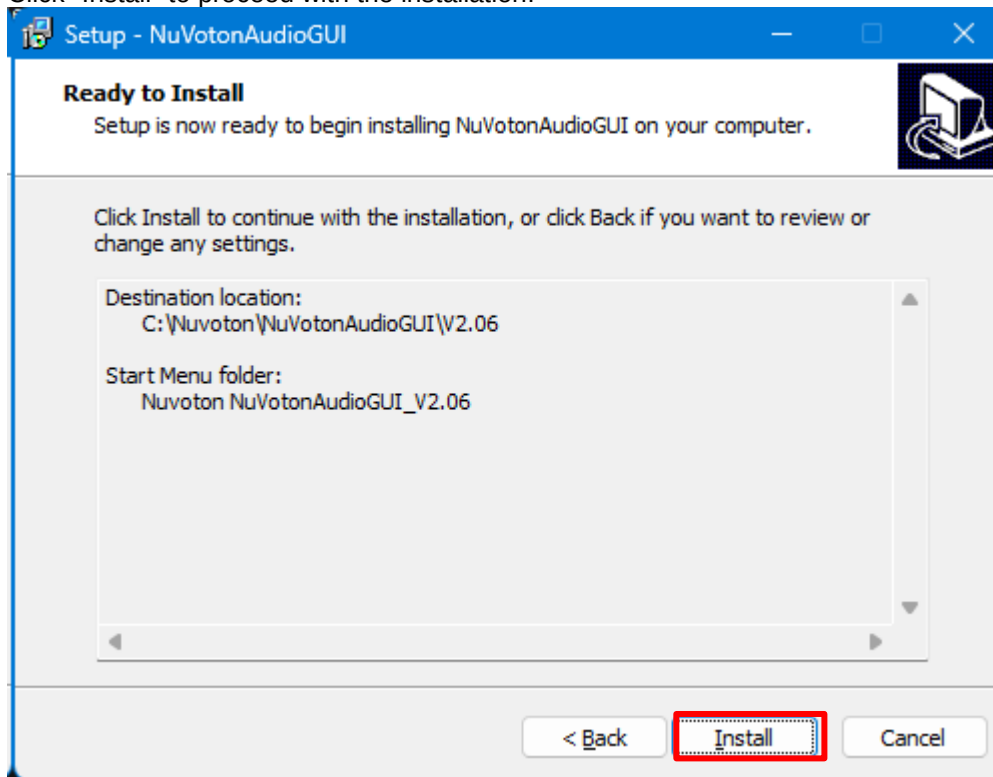
- Click "Next" to continue the installation.



- Click "Next" to continue the installation.

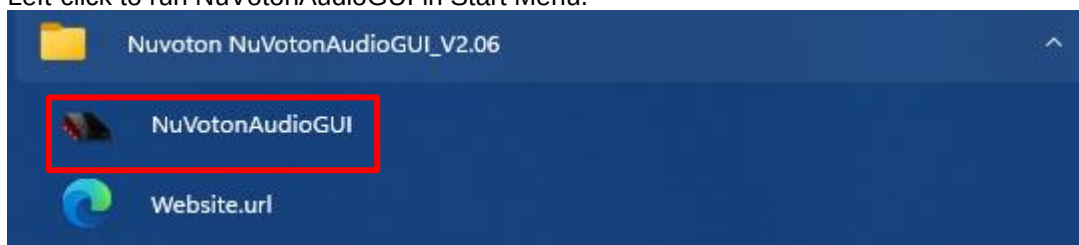


- Click "Install" to proceed with the installation.

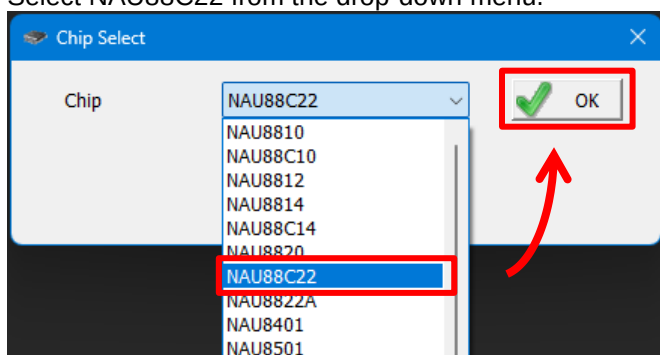


5 CONNECT TO GUI

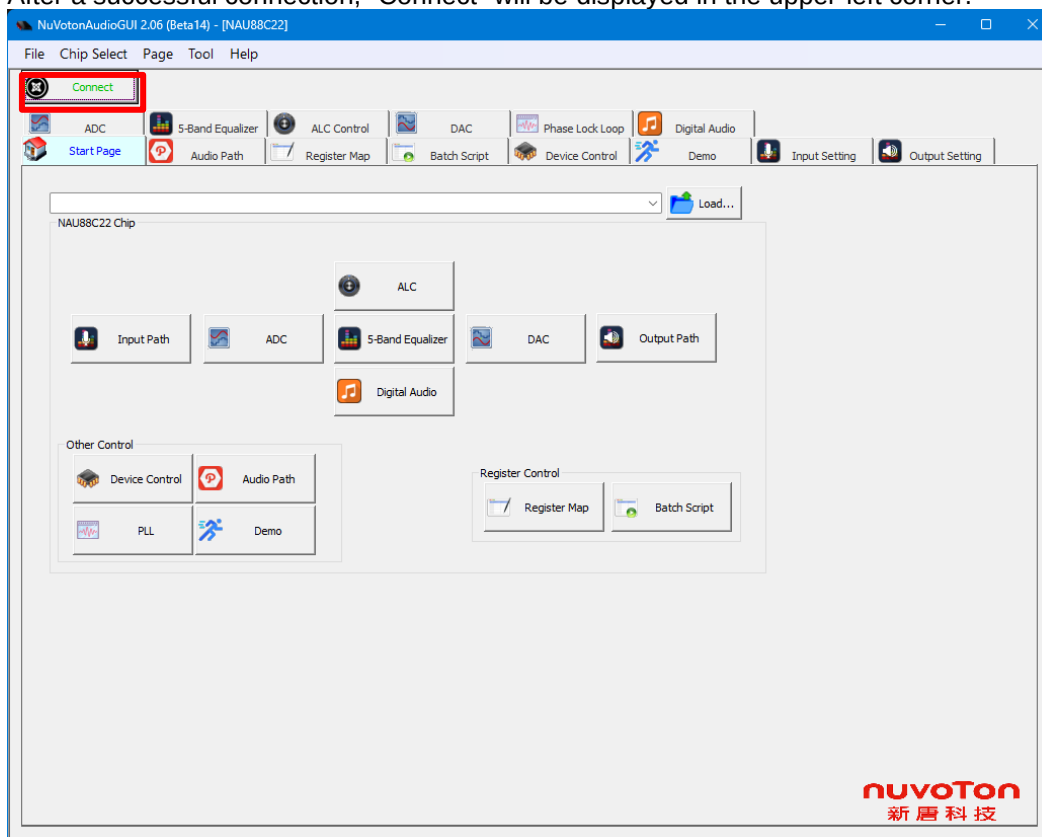
1. Left-click to run NuVotonAudioGUI in Start Menu.



2. Select NAU88C22 from the drop-down menu.

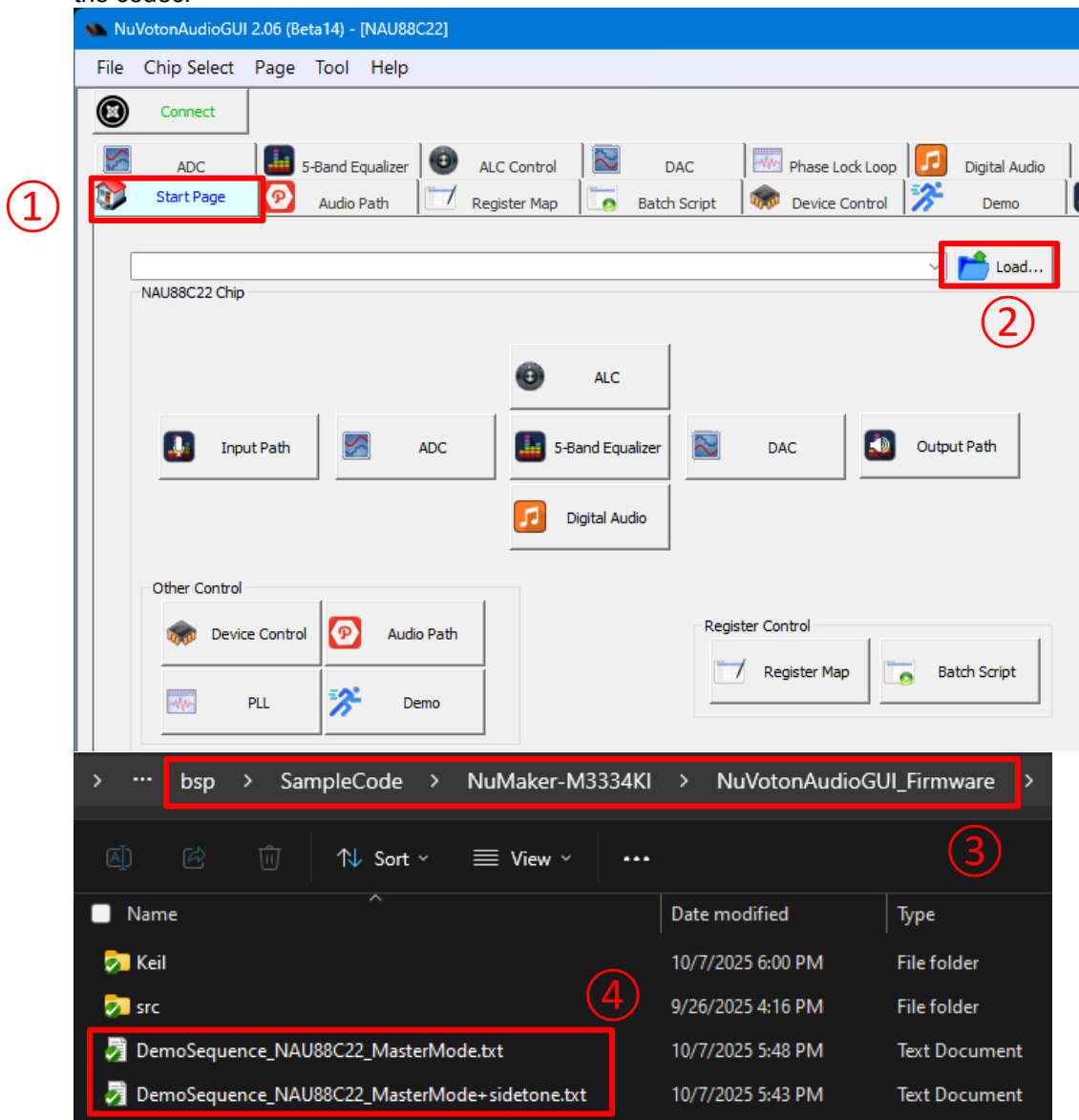


3. After a successful connection, "Connect" will be displayed in the upper-left corner.



6 OPERATE THE GUI TO CONFIGURE THE CODEC

1. **Start Page:** Import default or customized codec parameters to play/record audio files through the codec.

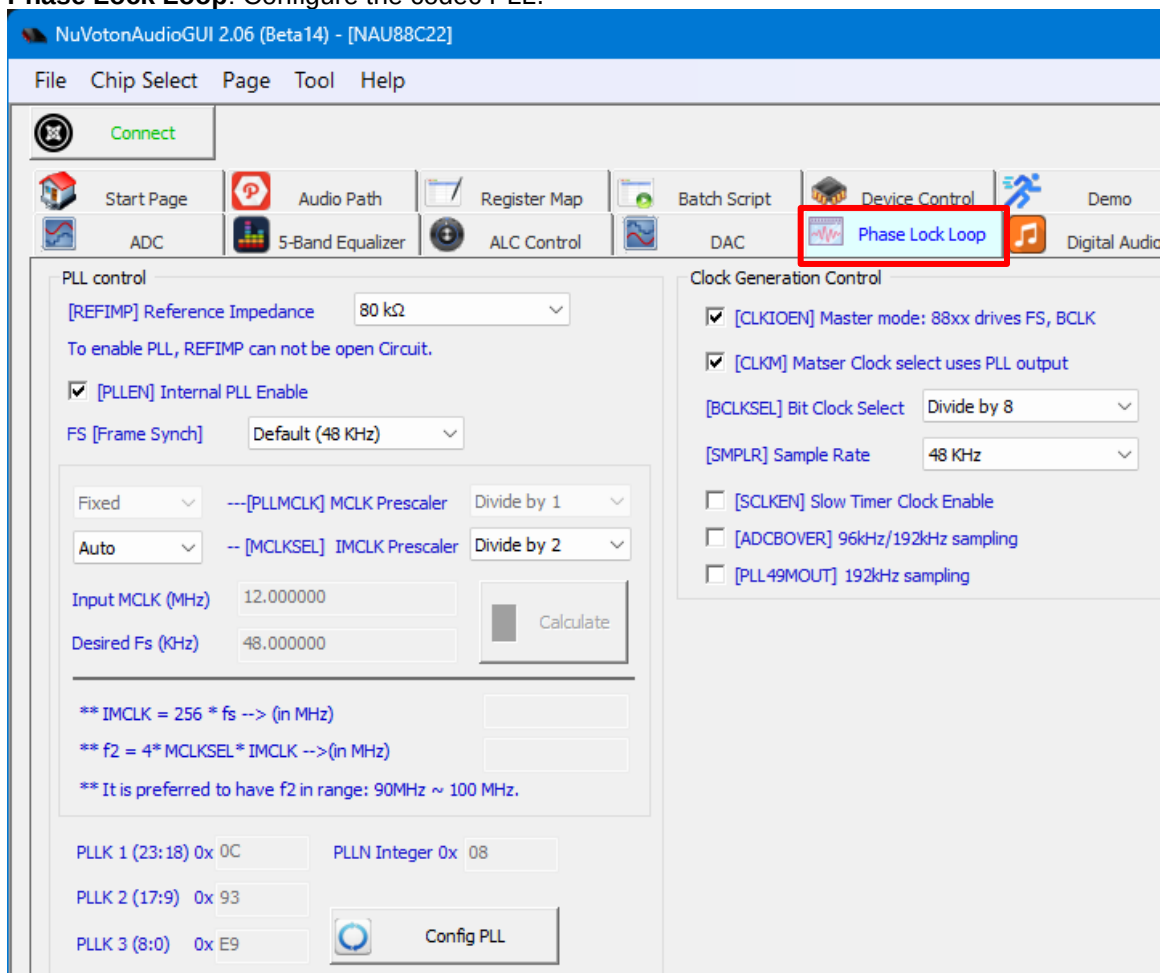


2. This sample code provides two sets of settings for users to choose from: DemoSequence_NAU88C22_MasterMode and DemoSequence_NAU88C22_MasterMode+sidetone.
 - DemoSequence_NAU88C22_MasterMode:
This setting enables the MIC IN (ADC function), SPK OUT (DAC function) functions of the NAU88C22 and sets the internal PLL oscillator output as master clock. Users can play their desired audio files using the playback device on the PC with the system playback device set to "Nuvoton UAC+HID Device". The audio will then be heard through the audio jack installed on NuMaker-M3334KI_V1.0.
 - DemoSequence_NAU88C22_MasterMode+sidetone:
This setting is almost identical to DemoSequence_NAU88C22_MasterMode, with the difference being that the speakers or headphone can directly play the sound received by the microphone.

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- NuVotonAudioGUI 2.06 (Beta14) - [NAU88C22]
- File Chip Select Page Tool Help
- Connect
- ADC 5-Band Equalizer ALC Control DAC Phase Lock Loop Digital Audio
- Start Page Audio Path **Register Map** Batch Script Device Control Demo
- Read All Read Address Value
- Write All Write 0x 00 0x 000
- Success
- Export... Import...
- Goto **Hint**
- Register Map Setting NAU88C22 Register Map
- ☐ Continue Read All Timer Interval: 1000 ms
- | Register Name | Address | Value(HEX) | Value(BIN) |
|-------------------|---------|------------|------------|
| SOFTWARE_RESET | 00 | 000 | 000000000 |
| POWER_MANAGEMENT1 | 01 | 12D | 100101101 |

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- The screenshot displays the NuAudioGUI 2.06 - [NAU88C22] application window. The top menu bar includes File, Chip Select, Page, Tool, and Help. Below the menu is a toolbar with icons for Connect, ALC Control, DAC, Phase Lock Loop, Digital Audio, Start Page, Audio Path (highlighted with a red box), Register Map, Batch Script, Device Control, Demo, Input Setting, Output Setting, ADC, and 5-Band Equalizer.
- The main workspace shows a detailed block diagram of the audio path. Inputs include LAUXIN, LMICN, LMICP, LLIN, RMICN, RMICP, RLIN, RAUXIN, VREF, and MICBIAS. These inputs feed into various processing blocks such as LADC MIX BOOST, RADAC MIX BOOST, HPF, ALC, Notch Filter, Limiter, LDAC, RDAC, LINMIX, RMAIN MIXER, AUX1 MIXER, AUX2 MIXER, and RSPK SUBMIXER. The outputs are labeled AUXOUT1, AUXOUT2, LHP, RHP, LSPKOUT, and RSPKOUT. Power supply pins like VDDA, VDDC, VSSD, VDDA, VSSA, VDDSPK, and VSSSPK are also indicated.
- On the right side, there is a Property panel with a tree view showing settings for different components:
- AUX1MIX2**: Not Connected
 - RSPK Sub-Mixer Setting**:
 - RAUXRSUBG: -15dB(default)
 - RSPKGAIN: 0.0 dB
 - RSPKMUT: Muted
 - RMIXMUT: Unmuted
 - RSUBBYP: Directly connect...
 - AUXOUT1 Setting**:
 - AUXOUT1EN: Off
 - AUXOUT1MT: Not Muted
 - AUX1HALF: Normal
 - AUX1BST: -1.0x gain
 - AUXOUT2 Setting**:
 - AUXOUT2EN: Off
 - AUXOUT2MT: Not Muted
 - AUX2BST: -1.0x gain
 - LHP Setting** (highlighted with a blue box):
 - LHPEN: On
 - LHPMUTE: Not Muted
 - LHPGAIN: 0.0 dB
 - RHP Setting** (highlighted with a blue box):
 - RHPEN: On
 - RHPMUTE: Not Muted
 - RHPGAIN: 0.0 dB
 - LSPK Setting**:
 - LSPKEN: On
 - LSPKMUTE: Not Muted
 - LSPKGAIN: 0.0 dB
 - SPKBST: -1.0x gain
 - RSPK Setting**:
 - RSPKEN: On
 - RSPKMUTE: Not Muted
- At the bottom of the interface, there are three status bars: PLL, AUDIO INTERFACE (PCM/IIS), and CONTROL INTERFACE (2-, 3- and 4-wire).

5. Phase Lock Loop: Configure the codec PLL.



REVISION HISTORY

Date	Revision	Description
2025.10.08	1.00	1. Initially version.

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